Quiz 1

Name: _____ Student ID: _____ Marks: 50

(1) Determine whether the sequence converges or diverges. If it converges, find the limit.

(a) $a_n = \sqrt{n+1} - \sqrt{n}$. [5]

(b) $a_n = n\cos\left(\frac{1}{n}\right)$. [5]

(c) $a_n = 7^{\frac{6n-1}{2n+5}}$. [5]

(2) Show that the sequence $\left\{\frac{5n+1}{6n+2}\right\}_{n=1}^{\infty}$ is monotonic. Find out whether it is increasing or decreasing. [5]

(3) Determine whether the series converges or diverges. You do not need to find the sum.

(a) $\sum_{n=1}^{\infty} \frac{1}{3+5^n}$. [5]

(b) $\sum_{n=1}^{\infty} \frac{n^3+1}{3n^3+4n^2+2}$. [5]

(c) $\sum_{n=1}^{\infty}(2^{\frac{1}{n}} - 1)$. [5]

(4) Find the radius of convergence and interval of convergence of the series [5]

$$\sum_{n=0}^{\infty} (-1)^n \frac{x^{n+2}}{2^{n+1}}.$$

(5) Find the power series representation of $\tan^{-1} x$. Use the result to show that [10]

$$\frac{\pi}{6} = \frac{1}{\sqrt{3}} - \frac{1}{3\left(\sqrt{3}\right)^3} + \frac{1}{5\left(\sqrt{3}\right)^5} - \frac{1}{7\left(\sqrt{3}\right)^7} + \frac{1}{9\left(\sqrt{3}\right)^9} - \frac{1}{11\left(\sqrt{3}\right)^{11}} + \cdots.$$